

Analysis of Seed Testing to Improve Cultivation using Image Processing Techniques

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Abstract—

This research study intends to smartly enhance the crop cultivation process by identifying the health status of the plant seed as defective or non-defective using the seed testing process. Seed testing aims to enhance agricultural initiatives, entails locating seeds and determining their defective status. This study has employed digital image processing technologies to test the purity of various seeds. Instead of performing physical purity analysis to calculate the proportion of pure seed components in a seed batch, here digital image processing techniques are used to accurately predict the defect from the seed picture. People can easily take digital photos with a camera or a mobile phone device with the recent advancements in camera technology.

Keywords—Image Processing, Real-Time Detection, Seed Testing, Seed Testing App, Seed Quality Check.

I. INTRODUCTION

1.1 Overview

This research study aims to introduce how image processing is used in seed technology, vigor techniques, and seed germination. Computer Vision (CV) can be used to distinguish the germination of seeds in very controlled situations. Image analysis and equalization are both used in the calculations for image skeletonization and image preparation. Changes to the seed's length and territory can be seen once the seed picture has been converted to binary. Techniques for computer vision are thought of as a continuous seed testing process. Computer technology helped professionals create frameworks for analyzing the rate length germination of seeds of rate length improvements [1]

Cameras and scanners are certainly used to capture images of seeds. Images of seeds that have already germinated are counted and evaluated using morphology operations and fuzzy logic techniques. Computer vision can monitor seeds and distinguish

between those that have and have not germinated. The seed germination test might benefit from these discoveries. These might aid researchers and the relevant sector in developing their capacity for testing seed and quality. To perform numerous processes for assessing and organizing seed quality, it is important to determine distinct physical and physiological features of seeds. Crop seed analysis gains more from image processing's low cost, fast speed, and ability to regulate environmental condition systems with computer hardware. According to techniques, and their H/W design, the machine image is appropriate for a seed biology laboratory. Technology for tracking seed ingestions has recently been developed using computer-aided techniques. The test is necessary to accurately reflect a seed's capacity for germination. Image recording was employed to determine the timing of radicle emergence. [2,3]

1.2 Image Processing

Deep learning has had a huge impact on many fields of technology in recent years. Computer vision, or the capacity for computers to know images and videos on their own, is one of the popular topics in this sector. Computer vision is required for biometrics, self-driving cars, and facial recognition to function. Image processing is at the heart of computer vision. Before we get started with image processing, we must first define what are the constituents of an image. The number of pixels is nothing but the dimensions of an image. For instance, if an image's xi dimensions are 200 x 500 (width x height), the overall number of pixels within the image is 100000. One such pixel is a spot on the image which gets on a particular colour, shade, or opacity. It is represented as either a grayscale image or RGB.

It is the process of doing some activities on an image to improve it. Image Processing is a type of signal

processing in which the system takes the image as input and the output can be image characteristics or features. A seed image takes the following three steps for processing.

They are listed below:

Adding the image to the system
Evaluating and modifying the image.
Output can be a report based on image analysis or an altered image.[4]

1.3 OPEN CV

Computer Vision is a process in which we can understand how images work, how we could store the images, how to make modifications to the image, and how we could extract as much amount of data as we can. Robotics and photo editing apps all heavily rely on computer vision. The shading-based approaches are the easiest way to recognize and separate an object from an image. To properly distinguish objects using shade-based techniques, the team, and the foundation need to have a significant difference in shading. To put it another way, acquired pictures may be thought of as three matrices, BGR (BLUE, GREEN, RED), with entries spanning from 0 to 255. These pixels are so small in real images that the human eye cannot distinguish between them [5]

The rest of the paper is organized as follows: Section II discusses the previous works related to seed testing in image processing. Section III explains the Materials and methodologies involved in seed testing to improve cultivation. The proposed system to detect seed testing is analyzed in section IV.

II. RELATED WORKS

Bala et al. [6] proposed the Protein-protein Interaction Prediction using Variable Length Patterns to create another framework to effectively equalized the steps of the Variable-Length patterns method utilizing a hybridized a priori and fp-growth on GPU; and to enhance performance indicators like computational time, memory use, and CPU utilization. Statistical Analysis and Methods: This study suggests using graphics processing units to speed up the prediction of protein-protein interactions (PPIs) (GPUs). For the same price as a CPU, a GPU can offer more cores and computational capacity. Findings: PPIs can be predicted using the commonly occurring patterns found in protein sequences. Although switching from fixed length to variable length techniques increased computational complexity, it was also shown to be beneficial. Applications and Improvements: Several computer engineers are currently studying sequence biology.

Balachandran et al. [7] stated that in the passive wireless detection of ethylene gas, a patch antenna is coupled with a SnO₂ capacitive sensor. In post-harvest handling, there is an annual loss of fruits and vegetables of roughly 25%. This can be avoided by regularly checking the state of fruits and vegetables and wirelessly transmitting the data without manual examination. At various ripening phases, climatic fruits like apples and peaches often generate varying amounts of ethylene. The

condition of the fruits can be predicted based on the levels of ethylene gas released. There is a lot of use of ethylene sensors for this.

Belsare et al. [8] said that Utilizing Image Processing to Assess the Quality of seed. The strategies utilized in seed technology that can be effectively implemented utilizing systems are reviewed in this research. The methods covered in this study include measures of germination index, seed vigour testing, and the seedling growth rate assessment. To track seed germination and vigour, computer-aided image analysis approaches have recently been developed. Because it has been too time-consuming and labor-intensive to consistently monitor seedling growth over time, although it is challenging to include in routine vigour testing, the seedling growth rate is thought to be a sensitive indicator of seed vigour. This study examines various computer-assisted methods that decrease the amount of manual effort needed to gauge seedling development rate and improve the precision of these readings. Also briefly given are a few seed-testing theories.

Beri and Singh [9] proposed the work that Web users routinely visit a wide range of websites as part of their daily life. The usability and conversion rates of a website are proxies for the user experience that users will have there. The usability and conversion rate of a website affects its success. A website's usability and conversion rate provides insight into its efficiency. Web analytics refers to a range of metrics and techniques that allow webmasters to assess the usability and conversion rate of their websites. This article aims to investigate how web analytics could raise website conversion rates and usability.

According to Chaugule [10] stated the purpose of this work is to provide an overview of seed technology, including image processing-based approaches to seed germination and vigor. Recently, techniques for computer-assisted image analysis were created to keep track of the growth and wellness of seedlings. They must be combined with the conventional germination test to appropriately depict the germination performance of a seed sample.

Rode et al. [11] explored the availability of high-quality seeds and the maintenance of seed standards have become critical challenges since the seed is an essential component for farmers, growers, and producers. Seed testing is the process of characterizing the physiological, genetic, and physical characteristics of seeds to assist informed decisions during research and development, seed production, along supply chains, and in commerce. The parameters used in seed quality testing are described in depth in the following presentation.

Sako et al. [12] stated that Seed vigour testing is a helpful tool for determining the quality of seed lots. This study offers a dependable, inexpensive, and user-friendly automated seed vigour testing system. The system links a computer with a digital imaging gear that captures images

of seedlings as they emerge. A computer processes the photos to provide values that represent the quality of a seed lot using some statistics derived from the morphological traits of the scanned seedlings. The sample averages of the hypocotyl and radicle lengths, as well as the sample standard deviation of the total length and the ratio of radicle-to-hypocotyl-length, are the statistics that demonstrate the rate and regularity of seedling development. The results demonstrated that the imaging method delivered trustworthy, unbiased results.

Li et al. [13] present a novel application of computer vision algorithms to detect lettuce seed germination in a controlled environment. In this application, there are two processes: image acquisition and analysis which makes use of image processing and image skeletonization techniques. After the conversion of seed images to binary images, the seed skeletons are extracted and used to identify changes in seed length. Estimates are also made for the area and other factors. The area and length of the seed are monitored when the radicle emerges to ascertain when germination starts. The results of our approach also suggest that when combined with germination.

III. PROPOSED SYSTEM

The proposed system sequence of steps is illustrated in detail in fig.3.1.

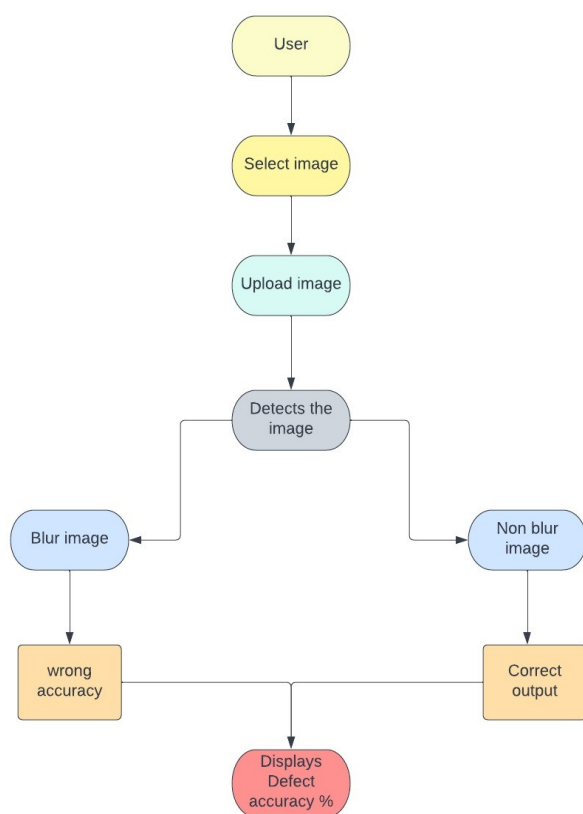


Fig. 1. Proposed System Architecture

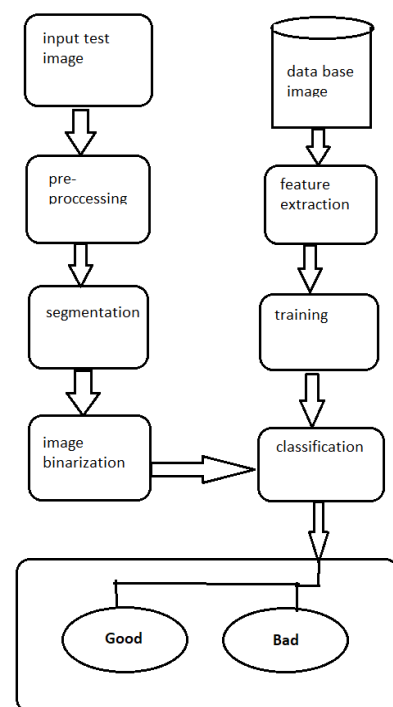
Fig. 1 shows the steps involved in seed testing, First, we have to capture the image and select which image we have to test and upload it, the system can detect the image, if the image is blurred it shows wrong accuracy,

and if not, it shows correct accuracy and displays whether the seed is a defect or not.

3.1 Algorithms Used:

The below steps and Fig.2 are described to detect seed tests to improve cultivation.

1. Image acquisition (input image) camera captures the image of the grain, and we get the input image.
2. Pre-processing and segmentation we convert RGB to gray and the background is eliminated.
3. Image binarization of an image based on its contrast levels.
4. The database collects images from multiple sources.
5. Extract the features from the image like area, major axis, minor axis, and perimeter.
6. Training /classifier is used to train the collective data and classifies the images according to features. in this case, the classes are good and bad.



IV. RESULTS AND DISCUSSION

4.1 Data Set Used

We trained 45 wheat images 12 single bad seeds, 19 groups of bad seeds, and 14 mixed seeds, out of 59 corn images 20 were bad, 21 groups of bad seeds and 18 mixed images, and rice 30 bad seeds, 16 group images, and 21 mixed images [14-18].

We trained these images by using the teachable machine. the trained images are taken from different sources from different sites.

Table1. Description of Database Images

Bad seeds	Number of images		
	single bad seed	group of bad seed	mixed seeds
Wheat	12	19	14
Corn	20	21	18
Rice	30	16	21

Table 1 shows that trained the bad seeds and mixed seeds with different images.

4.2 Implementation Results

Table 2 summarized the results and accuracy of the seed test cultivation.

Table 2. Conclusion of the Results and Accuracy

Bad Seeds	No.of database images	No.of images match	No.of images not match	Detection accuracy
Wheat	45	4	41	90 %
Corn	59	6	53	91.4 %
Rice	67	58	9	14 %

We can conclude these results are:

In the database of 45 images, the algorithm detects the wheat seeds in 4 images correctly, these correct images are bad seeds, and the remaining 41 images are good seeds so the accuracy is 90%.

In 59 images, the algorithm detects 6 bad images, and 53 good images so good seed accuracy is 91.4%.

From 67 images it detects 58 are bad seeds and only 9 are good seeds so good seed accuracy is 14%.

Table 3. Checking Results

Sample	Actual Number	Good seed	Result
Wheat	200	200	0 %
Corn	130	128	0.17 %
Rice	178	85	49.7 %

Based on seeds the abole3 shows accurate results, above we provide 200 good seeds for 200 seeds, there are no bad seed seeds, so the error percentage is 0%. Similarly, for corn, only 2 bad seeds we provide the error rate is 0.17%. finally, for rice, we provide 85 good seeds out of 178 seeds so the error rate for bad seeds is 49.7 because of more bad seeds.

Fig.2, the bar chart represents the results of bad seeds identification by the proposed algorithm.

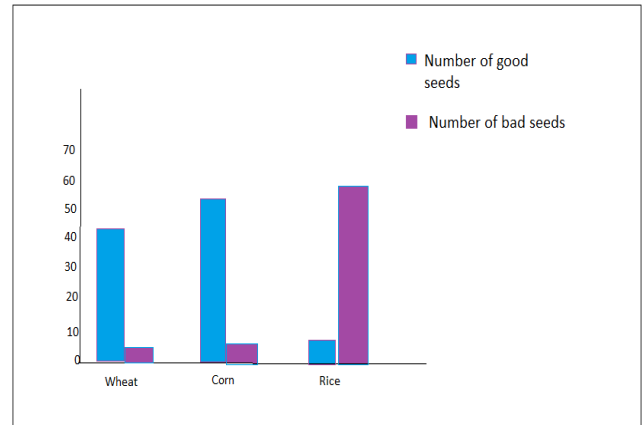


Fig.2. Identification of Bad Seeds

4.3 Steps to execute Proposed Work

Step 1: Open Android Studio

Step2: Choose a virtual device or Physical device from the menu.

Step3: Click on the project Run.

Step 4: View the application performance on a virtual or physical device.

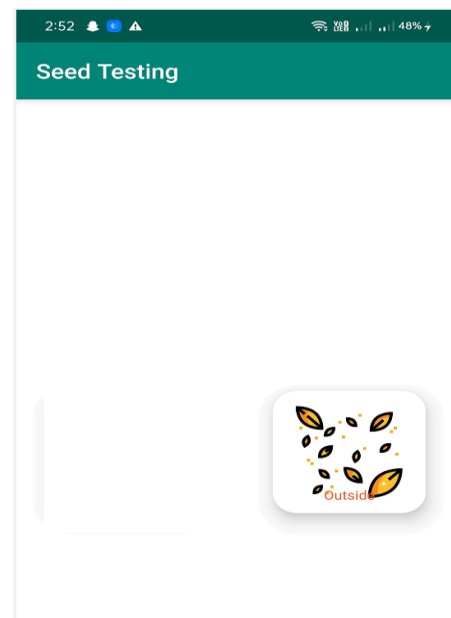


Fig.3 Application Interface

The above Fig.3 shows the User interface of the application, here we can see the option for opening the seed detection.

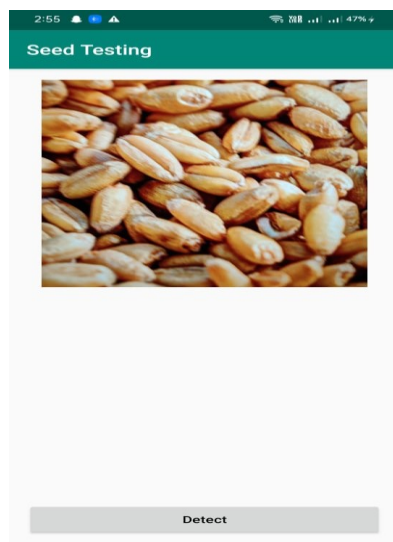


Fig 4. Testing Interface

Fig 4. shows the detect option for seed detection, on click detects the seeds and performs internal implementation.

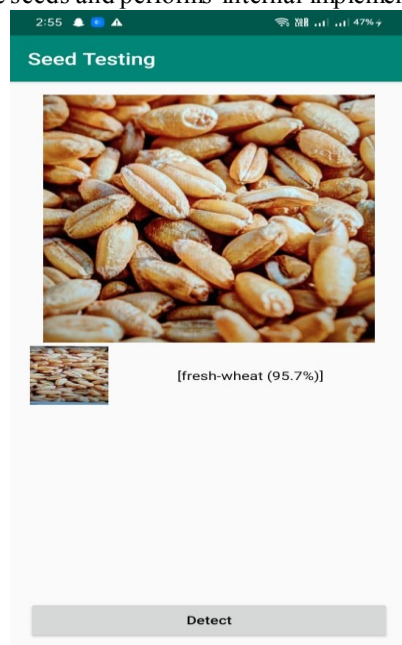


Fig. 5 Result Interface

Fig 5. shows the accuracy results of seeds, whether it is good or bad.

V. CONCLUSION AND FUTURE WORK

The proposed research work concludes by stating that even Farmers can use it to quickly determine whether the seed is damaged. The TensorFlow model, which has been trained on various seeds, is used by the application to detect the seeds. The name of the seed is also detected by this application in addition to whether it is damaged or not.

Seed testing is a crucial stage in assessing the quality and viability of seeds. Accurate testing allows growers to choose the right seeds for their crops, resulting in more successful harvests and higher yields.

For farmers and gardeners, seed testing can offer useful information that can guide them in selecting the finest seeds for their crops. Additionally, it can give details on

how the seed should be handled and stored, as well as an estimate of how many seeds need to be planted.

By using image processing we detect whether the seed quality is good or bad. for future scope, implement the seed information based on data, which is suitable soil, water needs a season to grow.

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